

STREAMSENTINEL

6 agents. 45 seconds. Here's what travels between them.

Most engineers talk about what each agent does.

This is about how they *talk to each other*.

THE PROBLEM

**Most multi-agent systems
are built as a feature list.**

✗ Agent 1 detects the problem

✗ Agent 2 diagnoses it

✗ Agent 3 decides what to do

Nobody thinks about what **data travels between them**.
That's where systems break. The handoff *is* the architecture.

HANDOFF 1

WatcherAgent

detects anomaly

**DiagnosisAgent**

explains why

PAYLOAD THAT TRAVELS

- anomaly type + severity
- raw transaction data
- timestamp + topic
- similar past incidents from memory

MemoryAgent enriches this payload *before* the handoff —
so DiagnosisAgent never starts cold.

HANDOFF 2

DiagnosisAgent

LLM root cause

**BlastRadiusAgent**

scores impact

PAYLOAD THAT TRAVELS

- full AI diagnosis
- root cause + business impact
- original anomaly context

BlastRadiusAgent uses **zero AI** — pure BFS graph traversal.
Governance decisions can't be probabilistic.

HANDOFF 3 – THE GOVERNANCE GATE

BlastRadiusAgent

LOW / MED / HIGH

**RemediationAgent**

acts or escalates

PAYLOAD THAT TRAVELS

- blast radius score
- affected services list
- high criticality services
- autonomous action decision

LOW / MED

auto-fix silently

HIGH

page human · touch nothing

HANDOFF 4

RemediationAgent

takes action

**NarratorAgent**

writes post-mortem

PAYLOAD THAT TRAVELS – EVERYTHING

- original anomaly
- AI diagnosis
- blast radius assessment
- action taken + timestamp

NarratorAgent gets the full incident context.

Post-mortem writes itself. No human input. No 3am writing.

THE MEMORY LAYER

MemoryAgent doesn't sit in the chain.

It sits **across** all of it.

STORES

Every incident as a vector in pgvector

RETRIEVES

Similar past incidents before Handoff 1

LEARNS

Gets smarter with every incident
processed

PRIVATE

Local Ollama. Zero data leaves the
machine.

Prior context beats better prompting. Every time.

THE LESSON

Building agents is the easy part.
Designing what travels
between them is the job.

Every handoff is a contract.

Define it explicitly — or debug it forever.

What's the hardest handoff you've designed in a distributed system? □

github.com/Neelam95/StreamSentinel

#Kafka #SoftwareEngineering #BuildingInPublic